



INDIVIDUAL QUESTIONS – INTRODUCTION

CHAPTER CONTENTS.

Because the output is a structured network, we can apply a range of queries to explore the data. This gives us a library of **pre-existing approaches** to ask **practical questions** about the causal landscape described by the participants.

Ex

See also

- [Formatting your map for what you want to show](#) for the visualisation side.
- [Vignettes](#) for textual outputs.

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Individual views – How does the system work according to individual sources ?

This question asks about **individual perspectives** on mechanisms and outcomes. By viewing the results for each source in turn you can identify sources reporting specific and unusual pathways of change and analyse individual viewpoints in depth.

Summarising – How do the sources claim that the system works, in summary ?

Creating **overall maps** provides a **top-level view** of what your sources report about the system.

Vignettes – What is a typical source and what is their story ?

Writing the story for a typical source helps to create an engaging 'human' story, bringing your data to life, providing a **tangible example** that represents the broader trends in your dataset.

What are the narratives behind a specific link ?

It is important to frequently return to the original quotes associated with each factor or link to **understand how different stakeholders interpret and talk about key concepts**. This can be done in Causal Map by clicking on a link in the interactive map, or by printing out quotes for a particular filter (e.g. just for a single bundle of links) with additional context and metadata.

Which factors and links were most frequently mentioned ?

Similar to overall frequency counts in non-causal coding ("which themes are mentioned most often?"), we can count how often particular causal factors are mentioned, and the number of sources mentioning a factor.

Which factors and links are mentioned by the most sources ?

The number of citations can be a useful measure of importance, but it can be distorted if one source mentions some factor a lot of times. A useful alternative is to count the number of *sources* that mentioned a factor or link at least once.

Main outcomes. Which factors are mentioned most often as outcomes ?

What do we even mean by “outcome”? A factor might have a lot of incoming links, so it is often mentioned as the consequence of something, but it might also have a lot of outgoing links, so perhaps plays more of an intermediate role. One way to answer this question is to calculate the “outcomeness” for each factor: the number of citations of its incoming links divided by the total number of times it was cited.

Main drivers. Which factors are mentioned most often as drivers ?

Just as we can identify the factors which are in an absolute or relative sense most frequently mentioned as outcomes, we can do almost the same to identify key drivers:

- Identify those with the highest number of outgoing links
- Or identify those with the lowest outcomeness score

Splitting by groups. Are different groups involved in different ways ?

The simplest way to compare groups is to make separate causal maps for each group (e.g. men/women, project A/B, or by age group), and visually compare them. This allows you to identify **common patterns and context-specific factors**.

Comparing groups – What factors or links were mentioned more by some groups than others, in the same map ?

We can directly **compare groups** to find factors or links mentioned more by one group than another using a statistical test to find the most surprising differences between groups, taking into account the underlying frequencies.

Identifying groups – Are there different subgroups within the data ?

Where there are many sources, it may also be useful to identify which sub-groups of sources are substantially different from one another in terms of the stories they tell.

What are the emerging or unexpected factors ?

One way to do identify emerging or unexpected factors is to use the elements from your theory of change as your codebook while coding and only adding other elements when necessary, making a note of these additional elements.

Does the evidence support your theory of change ?

One way to do this is to use the elements from your theory of change as your codebook and only add other elements when necessary. Validated pathways of change (by showing which mechanisms are observed on the ground) and find gaps where expected pathways might be missing or where stakeholders list elements were not anticipated in the theory of change?

Assessing systems change

One of the most exciting applications of causal mapping is to assess change over time within a system. If we apply a systematic approach to coding (using blindfolded manual coding or AI-supported coding) we can compare the frequencies with which links or factors are mentioned over time. This becomes particularly interesting when applying inductive coding, so that new and emerging phenomena can be included into the codebook. Re-applying new codes to previously coded data would be very tedious with manual coding but is easy to do with AI-supported coding: [[860 AI extensions -- Clustering, Soft Recode, Magnetisation]]

Sentiment – Which changes are perceived as most positive or negative ?

Building on the previous example, you can analyse the sentiment in particular of the immediate impacts of your project or intervention.

Focusing on specific factors. What influences and outcomes are connected to a specific factor ?

Focus on a particular **element of your project** to understand the **direct and indirect influences leading to a specific factor** and all **outcomes it contributes to**. This helps in exploring the role of the factor as both an outcome and an influence within the causal system.

Looking downstream. What are the direct and indirect consequences of one or more factors ?

Map out all **direct and indirect paths** flowing from selected factors to:

- Understand both intended and unintended outcomes
- Identify feedback loops and circular relationships
- Understand the causal pathways from a specific factor and how it relates to other causal links and factors

Looking upstream. What are the direct and indirect influences on one or more factors ?

Looking upstream is just the opposite of looking downstream.

Path tracing – How do one or more causes affect one or more effects, including indirect pathways ?

What are the main causal pathways from an intervention to an outcome? We can trace **chains of influence** from a starting point like an intervention to a key outcome, revealing the step-by-step or branching logic described by the sources. We can even compare the strength of evidence for different pathways.

Source tracing – What are the consequences of one or more factors, looking only at stories told in their entirety by individual sources ?

Source tracing is more conservative than path tracing and helps us avoid [[The transitivity trap]].

Robustness – How robust is the evidence for that X influences Y ?

How robust is the network of evidence for the influence of one or more "driver" factors on another set of "outcome" factors?

Counting and comparing influences

How much evidence is there for the influence of our intervention on a valued outcome? Is that a lot? Can we compare these numbers across pathways?

Properties of the causal map – Which factors are reported as being causally central or causally peripheral ?

Because we can understand the map as a network, we can ask network-style questions. For example we can look at measures like "betweenness" to identify factors which are central in the network.

Properties of the causal map – What is the overall structure of the network ?

There is a host of other analytical lenses available, for example, are there relatively isolated islands within the network? Are there sub-systems which operate more on their own? How sparse or dense is it?

Properties of the causal map – Are there leverage points ?

Systems thinking gives us ways to identify possible leverage points within a network.

Properties of the causal map – Are there feedback loops ?

Can we identify positive feedback loops within the network which might serve to maintain or exacerbate phenomena?

Combining questions

Causal mapping gets really useful when you start to combine the different questions you might want to ask in order to answer more sophisticated questions. We can think of many of the techniques as filters which filter the view in a particular way. Using multiple filters allows you to build up an answer to a question. Usually, order matters.

Tribes. The most relevantly different subgroups in your data (by causal story)

The **Tribes** filter answers a very specific analysis question:

Showing group data as custom link labels on the map with optional significance test

Custom Links Label can use **any field that exists in the current links table**.

Names of tables and fields

We can think of a causal map as a database consisting of two tables, the links table and the sources table. We don't need to have a separate table for the factors because the factors can be derived from the links table.

Hierarchical coding

See also: [\[\[005 Minimalist coding for causal mapping \(\(minimalist\)\)\]\]](#); [\[\[015 Combining opposites, sentiment\]\]](#); [\[\[006 A formalisation of causal mapping\]\]](#).
